Contribution Report

Machine Learning – Final Project

Ensemble Methods: Boosting vs. Bagging

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Individual Contributions

Member	Modules/Code	Report Sections	Experiments	Slides	Other
Aytan Mardaliyev	Decision Tree (CART, Gini/entropy, weighted splits, feature importances, repr, unit tests)	Methods: Decision Tree	Baseline sanity-check vs. sklearn	Tree methods overview	Critical-path completion; interface stability
Ulker Rahimli	AdaBoost (SAMME, weight updates, α computation, staged_predict, stump wrapper)	Methods: AdaBoost	Scaling (1–200 stumps), noise robustness for AdaBoost, bias–variance (boosting)	Boosting overview	Weighted Gini integration with tree
Madina Mammadli	Random Forest (bootstrap, OOB, max_features, multiprocessing, averaged probabilities)	Methods: Random Forest	RF scaling (n_estimators, depth), OOB vs test, noise robustness for RF, bias–variance (bagging)	Bagging overview	Parallelization; RF unit tests
Habil Huseynov	PCA, K-Means (elbow, multiple restarts), DBSCAN (k-distance, core/border labels, ARI)	Methods: Unsupervised; Results: unsupervised analysis	Scree/elbow/k-distance plots; 2D PCA projections; ARI computation	Unsupervised results	Cluster–class alignment discussion
Avaz Huseynov	Dataset selection; preprocessing (scaling, imbalance handling); run_all.py orchestration	Introduction, Experimental Setup, Results (all supervised), Discussion, Conclusion	Baseline, head-to-head (5-fold CV), noise robustness orchestration, bias–variance orchestration	Motivation, results synthesis, conclusions	Report assembly (LaTeX), slide deck skeleton, contribution report, repo hygiene, reproducibility